

I am preparing for my semester exam please provide answers in human written the answers is not long and not short, Answers only medium, subject related to Cyber law & Security and Answers in easy word. For 5-6 marks .

Sure. Send me the questions one by one (or all together), and I will provide:

- Medium-length answers (perfect for 5–6 marks)
- Human-written style
- Easy English words
- Cyber Law & Security subject related
- Exam-oriented answers with headings and points

You can send the next question now.



Q1. Discuss the design and implementation issues in firewalls. for 5 marks

Design and Implementation Issues in Firewalls

A firewall is a security system that controls incoming and outgoing network traffic. It protects a computer or network from unauthorized access. While designing and implementing a firewall, many important issues must be considered.

1. Security Policy

The first issue is deciding the security policy. The firewall should clearly define which traffic is allowed and which traffic is blocked. A weak policy can allow attackers to enter the network.

2. Firewall Placement

The firewall must be placed at the correct location in the network, usually between the internal network and the Internet. Wrong placement may leave some systems unprotected.

3. Performance and Speed

A firewall checks every data packet passing through it. If the firewall is not powerful enough, it can slow down the network performance. Therefore, speed and efficiency are important issues.

4. Authentication and Access Control

The firewall should support user authentication and proper access control. Only authorized users should be allowed to access network resources.

5. Updating and Maintenance

Firewalls need regular updates to protect against new cyber threats. Improper maintenance may create security weaknesses.

6. Logging and Monitoring

The firewall should keep logs of network activities. Monitoring helps administrators detect suspicious activities and cyber attacks quickly.

Conclusion

Proper design and implementation of a firewall are necessary for strong network security. A good firewall improves protection, controls network traffic, and reduces cyber threats.



Q.2 differentiate Virus, war and trojan Horse with examples.

Difference between Virus, Worm, and Trojan Horse

Basis	Virus	Worm	Trojan Horse
Meaning	A virus is a malicious program that attaches itself to a file or program.	A worm is a self-replicating malware that spreads automatically through networks.	A Trojan Horse is a harmful program that appears useful or genuine.
Need of Host File	It requires a host file or program to spread.	It does not need a host file.	It also does not need a host file.
Spreading Method	Spreads when infected files are shared or	Spreads automatically through networks and	Spreads when users install fake or attractive software.

Basis	Virus	Worm	Trojan Horse
	executed.	internet connections.	
Main Purpose	To damage or modify files and programs.	To consume system resources and spread rapidly.	To steal data, passwords, or give unauthorized access.
User Action Needed	Yes, user action is usually needed.	No, it spreads automatically.	Yes, users are tricked into installing it.
Speed of Spread	Slower than worms.	Very fast spreading.	Spread depends on user installation.

Examples

- **Virus:** Brain Virus, Melissa Virus
- **Worm:** Morris Worm, WannaCry Worm
- **Trojan Horse:** Zeus Trojan, Trojan Banker

Conclusion

Virus, worm, and Trojan Horse are all types of malware, but they differ in their method of spreading and purpose. Understanding these threats helps in improving cyber security and system protection.



Q.3 what is denial of service (DOS) attacks? explain with Example and its effect on the Network. for 5 marks

Denial of Service (DoS) Attack

A Denial of Service (DoS) attack is a cyber attack in which an attacker tries to make a computer, server, or network unavailable to users. The attacker sends a large amount of fake traffic or requests to the system, which overloads the network and stops normal services.

The main goal of a DoS attack is to disturb the working of a website or network so that genuine users cannot access it.

Example of DoS Attack

One common example is a **flood attack**. In this attack, the attacker sends thousands of requests to a web server in a very short time. The server becomes busy handling fake requests and cannot respond to real users.

For example, if an online shopping website receives huge fake traffic during a sale, the website may crash or become very slow.

Effects on the Network

1. Network Congestion

The network becomes overloaded with unnecessary traffic, reducing its performance.

2. Service Unavailability

Users cannot access websites, emails, or online services during the attack.

3. Slow System Performance

Servers and systems become slow because they are busy processing fake requests.

4. Financial Loss

Businesses may lose customers and money due to service interruption.

5. Damage to Reputation

Frequent attacks can reduce customer trust in the organization.

Conclusion

A DoS attack is a serious threat to network security because it affects the availability of services. Proper firewalls, traffic monitoring, and security tools are important to protect networks from such attacks.



Q.4 Explain the role of Digital signature and there requirements in ensuring documents security. for 5 marks

Role of Digital Signature and Its Requirements in Document Security

A digital signature is an electronic method used to verify the authenticity and integrity of a digital document or message. It works like a handwritten signature but provides more security in electronic communication.

Digital signatures are created using cryptographic techniques and are widely used in cyber security, online banking, e-commerce, and legal documents.

Role of Digital Signature in Document Security

1. Authentication

A digital signature verifies the identity of the sender. It confirms that the document was sent by the actual person or organization.

2. Integrity

It ensures that the document has not been changed or modified after signing. If any changes are made, the signature becomes invalid.

3. Non-Repudiation

The sender cannot deny sending the signed document because the signature is uniquely linked to them.

4. Data Security

Digital signatures protect sensitive information from unauthorized access and tampering.

5. Legal Validity

In many countries, digitally signed documents are legally accepted and used in online transactions and government services.

Requirements of Digital Signature

1. Private and Public Key

Digital signatures require a pair of cryptographic keys:

- Private key for signing
- Public key for verification

2. Certificate Authority (CA)

A trusted Certificate Authority issues digital certificates to verify the identity of users.

3. Hash Function

A hash function is used to convert the document into a fixed-size code for secure verification.

4. Secure Algorithm

Strong encryption algorithms are needed to provide proper security and prevent forgery.

Conclusion

Digital signatures play an important role in securing electronic documents by ensuring authenticity, integrity, and security. They are an essential part of modern cyber law and electronic communication systems.



Q. 5 How do Authentication mechanism enhance the security of VPN (types and advantages)? for 5 marks

Authentication Mechanism in VPN and Its Advantages

A Virtual Private Network (VPN) is a secure connection that allows users to access a private network through the Internet. Authentication mechanisms in VPN help verify the identity of users before granting access. This increases the security of the network and protects sensitive data.

Types of Authentication Mechanisms in VPN

1. Password-Based Authentication

In this method, users enter a username and password to access the VPN. It is simple and commonly used.

2. Two-Factor Authentication (2FA)

This method uses two verification steps, such as a password and OTP (One-Time Password). It provides stronger security.

3. Certificate-Based Authentication

Digital certificates are used to verify the identity of users or devices. It is more secure than passwords.

4. Biometric Authentication

This method uses fingerprints, face recognition, or iris scanning for authentication.

5. Token-Based Authentication

Users use a hardware or software token that generates temporary security codes for login.

Advantages of Authentication in VPN Security

1. Prevents Unauthorized Access

Only verified users can connect to the VPN network.

2. Protects Confidential Data

Authentication secures sensitive information from hackers and cyber attacks.

3. Improves Network Security

Strong authentication methods reduce the risk of data theft and unauthorized activities.

4. Ensures User Identity Verification

It confirms that the person accessing the VPN is genuine.

5. Supports Secure Remote Access

Employees can safely access company resources from remote locations.

Conclusion

Authentication mechanisms play an important role in VPN security by verifying users and protecting data. Strong authentication methods improve privacy, confidentiality, and overall network security.



Q.6 Explain information security governance and risk management. only for 5 marks

Information Security Governance and Risk Management

Information Security Governance refers to the system of rules, policies, and processes used to protect an organization's information and IT resources. It ensures that security activities support business goals and follow legal and ethical requirements.

Risk Management is the process of identifying, analyzing, and controlling security risks that may harm information systems or data.

Information Security Governance

1. Security Policies

Organizations create security policies and guidelines to protect data and systems.

2. Roles and Responsibilities

It defines the responsibilities of employees and management in maintaining security.

3. Compliance and Legal Requirements

Governance ensures that the organization follows cyber laws, standards, and regulations.

4. Monitoring and Control

Regular monitoring and auditing help maintain effective security practices.

Risk Management

1. Risk Identification

Finding possible threats such as hacking, malware, or data theft.

2. Risk Assessment

Analyzing how serious the risk is and its possible impact on the organization.

3. Risk Control

Applying security measures like firewalls, encryption, and backups to reduce risks.

4. Risk Monitoring

Continuously checking systems to identify new threats and vulnerabilities.

Conclusion

Information security governance and risk management help organizations protect their data, reduce cyber threats, and maintain secure and reliable information systems.



Q. 7 What is security Architecture ? Explain its importance in INformation Security? for 5 marks

Security Architecture and Its Importance in Information Security

Security Architecture is a structured framework used to design, implement, and manage the security of computer systems, networks, and information. It includes security policies, technologies, and controls that protect data and IT resources from cyber threats.

The main purpose of security architecture is to provide a secure environment for storing, processing, and sharing information.

Importance of Security Architecture in Information Security

1. Protection of Data

Security architecture protects sensitive and confidential information from unauthorized access, theft, and misuse.

2. Risk Reduction

It helps identify security risks and provides measures to reduce cyber attacks and system vulnerabilities.

3. Secure Network Communication

It ensures safe communication between systems and users through firewalls, encryption, and authentication methods.

4. Supports Business Continuity

Security architecture helps organizations continue operations even during cyber attacks or system failures.

5. Compliance with Laws and Standards

It helps organizations follow cyber laws, security standards, and government regulations.

6. Better Access Control

Only authorized users are allowed to access important systems and information.

Conclusion

Security architecture is an important part of information security because it provides a complete framework for protecting systems, networks, and data from cyber threats. It improves security, reliability, and trust in information systems.



Q.8 Describe the system of key in Cryptography .
Differentiate between Symmetric and assymetric key
systems. And their importance in securing communication.
For 5 marks

Key System in Cryptography

In cryptography, a key is a secret code used to encrypt and decrypt information. The key system helps protect data from unauthorized access during communication. Cryptography converts plain text into unreadable form called ciphertext, and the correct key is required to convert it back into readable form.

There are mainly two types of key systems:

1. Symmetric Key System
2. Asymmetric Key System

Difference between Symmetric and Asymmetric Key Systems

Basis	Symmetric Key System	Asymmetric Key System
Meaning	Uses the same key for encryption and decryption.	Uses two different keys: public key and private key.
Speed	Faster in operation.	Slower than symmetric encryption.
Security	Less secure for key sharing.	More secure because keys are separate.
Number of Keys	Only one secret key is used.	Two keys are used.
Key Sharing	Secret key must be shared securely.	Public key can be shared openly.
Example	AES, DES	RSA, ECC

Importance in Securing Communication

1. Data Confidentiality

Cryptography protects information from unauthorized users during communication.

2. Secure Online Transactions

It secures banking, e-commerce, and online payment systems.

3. Authentication

Asymmetric systems help verify the identity of users and websites.

4. Data Integrity

Cryptographic keys ensure that data is not modified during transmission.

5. Privacy Protection

Sensitive information such as passwords and personal data remains protected.

Conclusion

Symmetric and asymmetric key systems are important methods in cryptography. They provide secure communication, protect confidential data, and improve overall cyber security in digital systems.



Q.9 Describe the components and functioning of electronic payments system. Explain How E-cash and credit/debit cards are used in such systems. for 5 marks

Electronic Payment System

An Electronic Payment System (EPS) is a method of making payments through electronic devices and the Internet without using physical cash. It is widely used in online shopping, banking, ticket booking, and digital transactions.

Components of Electronic Payment System

1. User or Customer

The person who makes the payment using digital methods.

2. Merchant

The seller or service provider who receives the payment.

3. Payment Gateway

A secure system that transfers payment information between customer and merchant.

4. Bank or Financial Institution

Banks process and verify transactions between customer and merchant accounts.

5. Communication Network

Internet or network connection used for transferring payment data securely.

Functioning of Electronic Payment System

1. The customer selects goods or services online.
2. Payment details are entered through a secure payment gateway.
3. The bank verifies the payment information.
4. If the transaction is approved, the amount is transferred to the merchant.
5. Both customer and merchant receive transaction confirmation.

Use of E-Cash in Electronic Payment System

E-cash is digital money stored electronically and used for online payments. Users can transfer e-cash through digital wallets or online banking applications. It provides fast and paperless transactions.

Example:

UPI wallets and digital wallets like Paytm or PhonePe.

Use of Credit/Debit Cards in Electronic Payment System

- **Credit Card:** Allows users to make payments using borrowed money from the bank, which is repaid later.
- **Debit Card:** Directly deducts money from the user's bank account during payment.

Cards are used by entering card number, CVV, and OTP for secure transactions.

Conclusion

Electronic payment systems provide fast, secure, and convenient digital transactions. E-cash and credit/debit cards are important tools that make online payments easy and efficient.

